

Ingress approach (nginx sidecar until APIM is in scope)

Runtime ingress decision — nginx sidecar vs alternatives

Decision point: Workshop 2 **W2-Q1** (NGINX vs APIM for routing/SSL) and **R-W2-6** (scope creep if bundled with Linux uplift).

Scope: Advisory — consultant documents options; **client platform approves and implements**. APIM **build** is out of scope unless change order.

Related:

1. Decision summary

Field	Value
Question	Should the Podman runtime VM use an nginx sidecar (Quadlet), host nginx , APIM-only ingress, or a hybrid (APIM + local proxy)?
Default in standards v0.1	nginx sidecar — sole published host port (8080); app containers internal only
Status	Open — pending dev service inventory and client platform sign-off
Risk if deferred	NSG rules, Terraform <code>apim_subnet_prefix</code> , and Quadlet templates may not match chosen path
Risk if bundled early	APIM workstream delays Linux uplift (charter boundary)

2. Context

Current state (dev)

- Shared Linux runtime host with **multiple Podman services** (Gorobi, ingestion, FastAPI APIs, Dagster).
- **NGINX** already used as reverse proxy (typically **host-level** today).
- **Token validation** duplicated in application code; initiative to centralize at **APIM** (e.g. Gorobi route).
- Deploy path **not fully automated**; dev assessment in progress ().

Target reference stack

Four HTTP workloads behind one published port — see :

Service	Internal port	Path
Gorobi	8010	/gorobi/
Ingestion	8020	/ingestion/
FastAPI APIs	8000	/api/
Dagster	3000	/dagster/

```
flowchart LR
    subgraph edge [Edge - separate workstream]
        FD[Front Door optional]
        APIM[APIM]
    end
    subgraph vm [Podman runtime VM]
        NGINX[nginx sidecar or host nginx :8080]
        G[gorobi]
        I[ingestion]
        F[fastapi-apis]
        D[dagster]
    end
    FD --> APIM
    APIM -->|APIM subnet only| NGINX
    NGINX --> G
    NGINX --> I
    NGINX --> F
    NGINX --> D
```

Network rule (standards): runtime NSG allows app inbound from **APIM subnet CIDR** (or internal consumers in dev) — not from the internet directly.

3. Options

ID	Pattern	Description
A	nginx sidecar (Quadlet)	Containerized nginx on Podman internal network; only nginx-proxy publishes 8080 . Prescriptive default in standards.
B	Host nginx	nginx package on RHEL; proxy to 127.0.0.1:8010 , etc. Same routing rules; not pull-deployed as OCI image.
C	APIM only	APIM reverse-proxies to each backend URL/port (or one URL if single service). No local proxy on VM.
D	APIM + nginx sidecar (hybrid)	APIM: TLS, JWT/API keys, products, external API surface. nginx sidecar: path routing across containers on one VM. Recommended target for multi-service runtime.
E	APIM + host nginx	Same as D but host nginx instead of sidecar — lower Quadlet complexity; less container-parity with apps.

4. Pros and cons

A — nginx sidecar (Quadlet) — standards default

Pros	Cons
Single published port — aligns with NSG “one ingress” model	Extra container to operate (health check, upgrades, memory ~64–128 MB)
App containers not exposed on host network — banking-friendly	SELinux volume mounts (:Z) and config paths add bootstrap complexity
Same pull-only deploy model as apps (ACR image + Quadlet)	Overkill if one HTTP service and host nginx already works in dev
Natural path routing for Gorobi, ingestion, /api/ , Dagster WebSocket	Does not replace APIM for centralized token validation
Consistent with reference IaC (podman_runtime_workload role)	Migration churn if dev already stable on host nginx
Rate limits and security headers at local edge	APIM still required for external auth if VM has no public IP

B — Host nginx on VM

Pros	Cons
Simplest ops for dev — familiar pattern, no extra container	Ingress config lives outside container lifecycle (Ansible vs image tag)
Easy debugging (<code>nginx -t</code> , host logs)	Weaker parity between dev/UAT/prod if prod uses sidecar
Lower resource overhead than sidecar	Still need APIM (or other edge) for centralized tokens
Same path-routing capability as sidecar	Golden image may need nginx package + hardening review
Fastest path to stabilize current dev	Pull-only runtime standard prefers OCI-managed ingress image

C — APIM only (no local nginx)

Pros	Cons
Managed reverse proxy + auth — matches SaaS preference	Multiple backends in APIM for each container/port on same VM (operational overhead)
Removes per-service token code (Gorobi first)	APIM is out of scope for v1 Linux uplift unless change order
Centralized analytics, throttling, developer portal	Extra network hop; latency vs on-box proxy
Can terminate TLS at edge without VM public IP	Dagster WebSocket and path-heavy local routing less ergonomic than nginx
Strong audit story for API access	Using APIM as dumb pass-through underuses the service
	NSG still needs defined source (APIM subnet)

D — APIM + nginx sidecar (hybrid)

Pros	Cons
Clear separation: APIM = security/governance; nginx = local fan-out	Two layers to design, test, and document
One APIM backend URL → VM :8080 ; nginx splits internally	Requires APIM subnet + <code>apim_subnet_prefix</code> in Terraform
Supports Gorobi token policy at APIM without nginx Lua	Two workstreams — must not block Linux uplift on APIM readiness
Matches reference architecture and workshop direction	Cost of APIM SKU + nginx container
Scales when adding services (new Quadlet + <code>location</code> block)	
Keeps runtime VM off the public internet	

E — APIM + host nginx

Pros	Cons
Hybrid benefits of D with lower container count	Split between host package and containerized apps
Practical dev → UAT bridge if dev already on host nginx	Standards drift if prod adopts sidecar later
APIM backend still single URL :8080	Same APIM out-of-scope boundary as C/D

5. APIM as reverse proxy (clarification)

Yes — APIM can act as a reverse proxy. It accepts HTTP(S), applies policies, and forwards to backends. That is not the same as replacing **on-VM** routing:

Concern	APIM	nginx (sidecar or host)
JWT / OAuth at edge	Primary fit	Possible but awkward
Route <code>/gorobi</code> , <code>/api/</code> on one VM IP	One backend URL; path rewrite policies possible but clunky for many local services	Primary fit
Multiple containers, one published port	Needs local proxy or many APIM backends	Primary fit
Dagster UI / WebSocket	Supported with limits; nginx often simpler locally	Primary fit
Cost / charter	Separate project / CO	In scope for Linux standards

6. Decision criteria

Use after **service inventory** (NEXT-STEPS 1.2) is complete.

Criterion	Favors sidecar / host nginx	Favors APIM-only (no local proxy)
HTTP services on one runtime VM	2+ services	1 public HTTP service
Need one NSG app port	Yes	Single service direct to APIM
Centralized token validation	APIM upstream (either way)	APIM
Dev stabilization priority	Host nginx (B) acceptable short-term	Wait for APIM before UAT
Container-parity deploy (pull-only)	Sidecar (A or D)	N/A
Charter / timeline	Local proxy in Linux uplift	APIM build deferred

7. Recommendation (advisory)

Phase	Ingress choice	Rationale
Dev (now)	B — host nginx if already working; do not block on sidecar migration	Mount/SELinux/deploy automation are higher priority
UAT / prod (Linux uplift v1)	D — APIM + nginx sidecar <i>when</i> multi-service stack confirmed	Matches standards, NSG model, and reference stack
UAT / prod (single API only)	APIM → single container port; sidecar optional	Avoid unnecessary container if inventory shows one HTTP surface
APIM workstream	Separate from OS uplift — Gorobi first route	W2-D2; out of scope unless CO

Default until client decides otherwise: retain **nginx sidecar** in and reference Ansible/Terraform; record explicit override in §9 sign-off if choosing B, C, or E.

8. Deliverables

8.1 Consultant (this engagement)

ID	Deliverable	Status	Target
ING-1	This decision document (options, pros/cons, criteria)	Done	Week 2–3
ING-2	Update standards ingress row if client selects non-default option	Pending sign-off	After ING-6
ING-3	NSG / allow-list alignment note in or	Pending	With network workshop
ING-4	Reference stack note in if paths/ports change	As needed	Phase 2

8.2 Client

ID	Deliverable	Owner	Blocks
ING-5	HTTP service inventory — which services need external ingress vs internal-only (Dagster daemon, batch, etc.)	Client dev lead	Decision
ING-6	Signed ingress choice (A–E) for dev vs UAT vs prod	Client dev lead / platform	ING-2, Terraform NSG
ING-7	Current nginx config export (host paths, <code>location</code> blocks, TLS)	Client dev lead	Dev migration plan
ING-8	APIM workstream charter (if C , D , or E) — Gorobi pilot route, token policy, subnet	Client dev lead	APIM implementation
ING-9	APIM subnet CIDR for firewall + <code>apim_subnet_prefix</code> tfvars	Network / Client platform	Runtime NSG lockdown

8.3 APIM workstream (out of scope unless change order)

ID	Deliverable	Owner	Notes
ING-10	APIM instance + VNet/internal mode (if required)	Client platform	Separate project
ING-11	API definitions: Gorobi, optional ingestion/API products	Client platform	Remove app-level token checks after pilot
ING-12	Front Door + WAF (optional)	Client platform	Pairs with APIM; not required for dev
ING-13	Policy: JWT validate, rate limit, backend = runtime :8080	Client platform	Consultant may review advisory

8.4 IaC / automation (client implements)

ID	Deliverable	Location	When
ING-14	Quadlet + nginx sidecar templates	ansible/layer2/roles/podman_runtime_workload/	Option A or D
ING-15	Host nginx Ansible role or cloud-init	New or compliance path	Option B or E
ING-16	runtime_app_ports / NSG APIM source CIDR	infra/terraform/modules/compute-linux/	UAT prep
ING-17	apim_subnet_prefix in workload tfvars		When APIM subnet known

9. Decision record (sign-off)

Field	Value
Chosen option (dev)	[A / B / C / D / E]
Chosen option (UAT/prod)	[A / B / C / D / E]
APIM workstream	[Separate CO / Deferred / N/A]
Effective date	[date]
Approver	[Client dev lead / platform]

10. Industry references

Full map:

Option	Source
nginx sidecar	nginx proxy module NIST SP 800-190 (network segmentation)
APIM	Azure API Management
API security	OWASP API Security Top 10
Banking edge	MCSB network security

11. Document history

Version	Date	Notes
0.1	2026-06-08	Initial decision point — nginx sidecar vs host nginx vs APIM; deliverables for W2-Q1